

NGUYEN MINH KHOA

AI Engineer & Researcher

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PROFILE

Applied AI engineer and Research Assistant at AiTA Lab, FPT University, working across model experimentation, data pipelines, and inference systems. Research spans handwritten mathematical expression recognition and text-to-molecule generation, with a current focus on AI for intelligent manufacturing.

RESEARCH EXPERIENCE

Research Assistant | AiTA Lab, FPT University

Oct 2024 - Present

- Developed and evaluated deep-learning methods across HMER and text-to-molecule generation, covering data preparation, model training, ablation studies, and reproducible experiment tracking.
- Selected published outcomes include up to **+5 ExpRate points** for Mask CoMER, **+4 ExpRate points** for MTL + CBARM, and **+3.4 BLEU** with approximately **30% lower Levenshtein distance** for ChemAligner-T5.
- 5 papers across three research topics:** HMER (*Mask CoMER*; *MTL + CBARM*), text-to-molecule generation (*LexiChem*; *ChemAligner-T5*), and active learning with vision-language models (*CorrTie*).

PROJECTS

LexiChem | AI & Inference Contributor

2025 - 2026

Capstone combining text-to-molecule research with a complete local demo for molecule generation and exploration.

- Worked in a 3-person capstone team; owned the AI data pipeline, SMILES/SELFIES conversion, method development, ablations, and the inference path between research models and the application backend.
- Combined L+M-24, Mol-Instructions, and ChEBI-20; used RDKit to parse structures, validate valence, canonicalize SMILES, remove duplicate molecules, and convert cleaned structures to SELFIES for training.
- Developed the shared-latent alignment method and ran controlled ablations. On ChEBI-20, improved BioT5+ from **87.2 to 88.8 BLEU**, **52.2 to 52.9 exact match**, and **90.7 to 93.5 MACCS FTS**, while maintaining **100% validity**.
- Wrote the paper Introduction and full Results & Discussion, and co-wrote Methods; presented LexiChem at APWeb-WAIM 2026 in Da Nang.
- Integrated the Experiments workflow with FastAPI and NVIDIA Triton, including prompt normalization, model dispatch, RDKit validation of returned SMILES, and experiment persistence in MongoDB.
- Worked with the team to productize generation into a local application with Dashboard, Experiments, Simulation, 2D/3D molecular exploration, and docking-oriented workflows. RAG/Knowledge remains in development.

NexOps | AI, Simulation & Platform Lead

2025 - Present

Local-first MES/SCADA prototype linking factory simulation, production workflows, dashboards, and real-device control.

- Led the AI, simulation, and platform work for a working local prototype covering machine telemetry, production planning, work orders, alarms, event history, and OEE by shift.
- Built repeatable machine and shift scenarios so planning, alarms, downtime, and OEE can be exercised through the same deterministic workflow during development and demos.
- Designed the telemetry path from simulator and edge events through MQTT/EMQX into FastAPI services, TimescaleDB, and operator-facing dashboard state.
- Connected the local runtime to production-facing views for machine status, planning, alarm handling, and OEE reporting rather than leaving the simulator as a standalone data generator.
- Integrated a Bambu Lab printer over LAN and exercised real-device control as a separate hardware path inside the prototype.

HMER Research | Research Contributor

2024 - 2026

Research track on converting handwritten mathematical expressions into structured LaTeX, with published work on pretraining and multi-task attention refinement.

- Contributed to *Mask CoMER*, a two-stage HMER approach using masked-language pretraining followed by task training with stochastic-depth regularization; published results improved ExpRate by up to **5 points** over the CoMER baseline.
- Contributed multi-task learning experiments for the *MTL + CBARM* line, where the main LaTeX-sequence task is trained with a structural auxiliary task and attention refinement; published results improved ExpRate by approximately **4 points** over the CoMER baseline.
- Trained and evaluated PyTorch/PyTorch Lightning models on CROHME benchmarks and tracked experimental runs with Weights & Biases.
- Worked across model experimentation, evaluation, and reproducibility rather than only manuscript preparation; both research lines resulted in published papers.

TrendRadar | Data Pipeline Owner

2025

AI-driven trend analytics platform for short-form content, built to turn heterogeneous TikTok, TikTok Shop, and Reels data into normalized records for downstream analytics and ML.

- Designed and implemented the end-to-end ingestion and preprocessing pipeline from data collection through normalization and PostgreSQL storage.
- Built asynchronous ingestion with Pydantic schema validation to enforce consistent contracts across heterogeneous sources.
- Added idempotent processing, deduplication, retry/backoff, and quarantine handling for malformed records to stabilize scheduled runs.
- Added pytest coverage, structured logging/monitoring hooks, and Docker-based local execution for reproducible pipeline development.
- Reduced ingestion failures by **30%** and authored **8 typed schemas** for source validation and normalization.

Research Ops | Developer

2026 - Present

Personal pre-alpha workspace for keeping sources, evidence, experiments, revisions, and human decisions traceable across long-running research cycles.

- Designed the project around reducing human attention required to keep trustworthy research moving while preserving provenance, checkpoints, and researcher judgment.
- Implemented a local research library, PDF/citation handling, work sessions, evidence links, manuscript exports, and resumable study state backed by FastAPI and PostgreSQL.
- Linked source files and cited spans through SHA-256 content digests so evidence remains tied to the exact source version used for a decision or manuscript revision.
- Added revision-aware exports and approval gates for runtime changes; the researcher retains control of direction, interpretation, and scientific conclusions while agents handle bounded execution.
- Current status is explicitly pre-alpha: no public deployment or measured long-term benefit yet, and model execution/GPU evaluation remain unfinished.

PUBLICATIONS

LexiChem: Shared-Latent Alignment for Faithful Text-to-Molecule Generation in SELFIES Space

APWeb-WAIM 2026 | Accepted and presented | **Scopus-indexed proceedings**

CorrTie: Correction-aware Tie-breaking for Active Learning with Vision-Language Models

APWeb-WAIM 2026 | Accepted | **Scopus-indexed proceedings**

Enhancing handwritten mathematical expression recognition with convolutional block attention refinement module and multi-task learning

Journal of Information and Telecommunication | **SCIImago Q2 (2025)** | Published online Aug 2026

ChemAligner-T5: A Unified Text-to-Molecule Model via Representation Alignment

ICCIES 2026 | Springer CCIS proceedings | Published

Mask CoMER: Enhancing Handwritten Mathematical Expression Recognition with Masked Language Pre-training and Regularization

ICDAR 2025 | **ICORE 2026 Rank A** | Published

EDUCATION

Bachelor of Science in Artificial Intelligence | FPT University, HCMC

2022 - 2026

Academic requirements completed; degree conferral expected Nov 2026.

AWARDS & SCHOLARSHIPS

- **50% Tuition Scholarship**, FPT University (2022), awarded through the university scholarship examination and covering 50% of tuition for the full programme.
- **RESFES HCMC Second Prize**, ChemAligner-T5 (2026).
- **RESFES HCMC Second Prize**, Mask CoMER (2025).
- **Demo Pitching Day**, VND 50M team award/project funding (2025).
- **Innovation Quest**, VND 40M team award/funding for TrendRadar / FrontAI (2025).
- **HCMC Open April Olympiad**, Silver Medal in Mathematics (2021).

TECHNICAL SKILLS

AI & Research: Python, PyTorch, PyTorch Lightning, Hugging Face Transformers, scikit-learn, OpenCV, RDKit, Weights & Biases, MLflow, self-supervised learning, vision-language models, cross-modal learning

Backend & Data: FastAPI, Pydantic, Pandas, PostgreSQL/TimescaleDB, MongoDB, Redis, MQTT/EMQX, REST, ETL/data pipeline design

Engineering & Reproducibility: Docker, Docker Compose, Linux, Git, Nginx, Playwright, LaTeX, scientific writing

CERTIFICATIONS

- **AIO 2024 | AI VIET NAM:** Machine Learning; Deep Learning; Computer Vision & NLP; GenAI & LLMs.
- **Machine Learning Specialization**, Stanford Online / DeepLearning.AI (2023).
- **Deep Learning Specialization**, DeepLearning.AI (2024).
- **Natural Language Processing Specialization**, DeepLearning.AI (2025).
- **DeepLearning.AI TensorFlow Developer Professional Certificate** (2025).
- **Big Data Specialization**, University of California San Diego (2024).
- **AI Foundations for Everyone Specialization**, IBM (2023).
- **Introduction to AI**, Google (2026).
- **IBM Full Stack Software Developer Professional Certificate** (2024).
- **AWS Cloud Technical Essentials**, Amazon Web Services (2025).
- **Software Development Lifecycle Specialization**, University of Minnesota (2024).
- **Research Methodologies**, Queen Mary University of London (2025).
- **Being a Researcher in Information Science and Technology**, Politecnico di Milano (2025).
- **Understanding Research Methods**, University of London / SOAS University of London (2025).
- **Introduction to Research for Essay Writing**, University of California, Irvine (2025).
- **Advanced Writing**, University of California, Irvine (2025).
- **Academic Skills for University Success Specialization**, The University of Sydney (2023).
- **Project Management Principles and Practices Specialization**, University of California, Irvine (2025).
- **Ethics in the Age of AI Specialization**, LearnQuest (2024).